

HLS based Rapid Pareto Front Search of Watermarked Convolutional Layer IP Design

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INTRODUCTION

- The exploration of design solutions for convolution layer hardware IP during HLS involves balancing two key parameters.
- Accurately identifying the optimal solution within a large design space and minimizing the time and resources needed for exploration.
- Efficient exploration methods are essential to simultaneously reduce evaluation time and maximize automation, addressing the diverse objectives of achieving the best architecture and speeding up the design space exploration process.
- This balance is crucial to managing the complexities of contemporary VLSI systems.

INTRODUCTION

- The HLS-driven Pareto set, or Pareto front, represents multi-objective optimal solutions that balance orthogonal constraints like design area and latency.
- Adding hardware security, such as watermarking, affects these constraints, making tradeoff analysis crucial to prevent IP piracy in VLSI design.

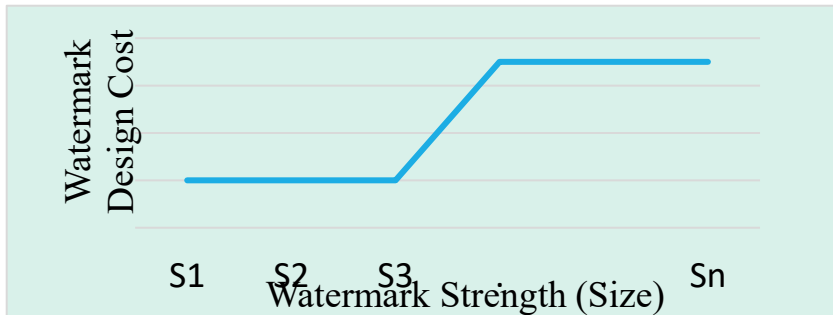


Fig. 1. Watermark design cost vs watermark strength

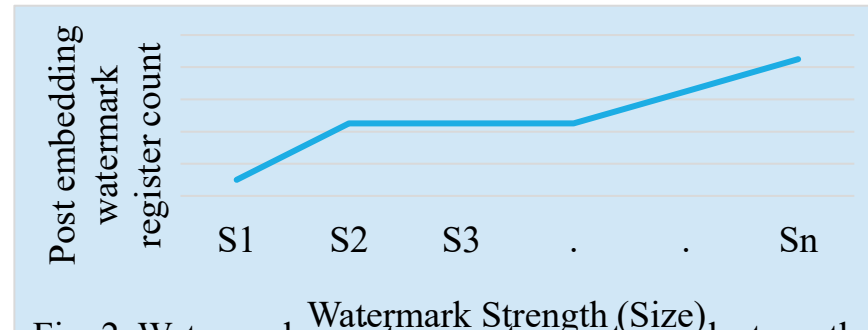


Fig. 2. Watermark register count vs watermark strength

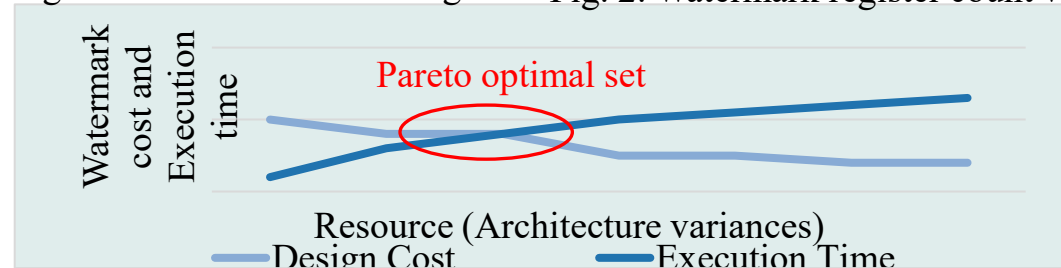


Fig. 3. Pareto front in design cost vs execution time tradeoff

PREVIOUS WORKS

Sr. No.	Existing Work	Technique Used	Remarks
1.	K. Vyas et al., [1] (2006)	using genetic algorithm	However, none of these approaches [1],[4] are capable to explore pareto fronts of HLS based watermarked IP designs, unlike proposed approach.
2.	Z. Zenget. al., [4] (2010)	using fuzzy searching	
3.	F. Koushanfar et. al., [2] (2005)	using behavioral synthesis techniques	without focusing on exploring optimal set of solutions that can perform watermark area/design cost vs execution time tradeoff.

PROPOSED METHODOLOGY

A. Watermark Design Cost parameter

- The watermark design cost of the watermarked IP design is determined using the eq.1

$$AR = (\alpha_{mul} * \beta_{mul} + \alpha_{adr} * \beta_{adr} + \alpha_{reg} * \beta_{reg}) + A(Rclk) \quad (1)$$

- Further, the total cost (C_R) of the resources using equation 1 is expressed as:

$$C_R = (\alpha_{mul} * \beta_{mul} + \alpha_{adr} * \beta_{adr} + \alpha_{reg} * \beta_{reg}) * \gamma + A(Rclk) * C_{Rclk} \quad (2)$$

- Applying partial derivatives to equation (2) with respect to α_{mul} , α_{adr} , and α_{reg} . Results:

$$\frac{dC_R}{d\alpha_{mul}} = \frac{d[(\alpha_{mul} * \beta_{mul} + \alpha_{adr} * \beta_{adr} + \alpha_{reg} * \beta_{reg}) * \gamma + A(Rclk) * C_{Rclk}]}{\alpha_{mul}} = \gamma * \beta_{mul} \quad (3)$$

$$\frac{dC_R}{d\alpha_{adr}} = \frac{d[(\alpha_{mul} * \beta_{mul} + \alpha_{adr} * \beta_{adr} + \alpha_{reg} * \beta_{reg}) * \gamma + A(Rclk) * C_{Rclk}]}{\alpha_{adr}} = \gamma * \beta_{adr} \quad (4)$$

$$\frac{dC_R}{d\alpha_{reg}} = \frac{d[(\alpha_{mul} * \beta_{mul} + \alpha_{adr} * \beta_{adr} + \alpha_{reg} * \beta_{reg}) * \gamma + A(Rclk) * C_{Rclk}]}{\alpha_{reg}} = \gamma * \beta_{reg} \quad (5)$$

$$\frac{dC_R}{dA_{Rclk}} = C_{Rclk} \quad (6)$$

PROPOSED METHODOLOGY

A. Watermark Design Cost parameter

- In accordance with the theory of differential approximation, the alteration in the overall area can be roughly estimated using the subsequent equation:

$$dC_R = \frac{dC_R}{d\alpha_{mul}} * \Delta\alpha_{mul} + \frac{dC_R}{d\alpha_{adr}} * \Delta\alpha_{adr} + \frac{dC_R}{d\alpha_{reg}} * \Delta\alpha_{reg} + \frac{dC_R}{dA_{Rclk}} * \Delta A_{Rclk} \quad (7)$$

- Substituting equations (3) to (6) into equation (7), yields equation (8):

$$dC_R = \Delta\alpha_{mul} * \gamma * \beta_{mul} + \Delta\alpha_{adr} * \gamma * \beta_{adr} + \Delta\alpha_{reg} * \gamma * \beta_{reg} + \Delta A(Rclk) * C_{Rclk} \quad (8)$$

- The dominance factors (DFs) of the resources wrt. watermark design cost are defined from eqns. (9) – (12).

$$DF(mul) = \frac{\Delta\alpha_{mul} * \beta_{mul} * \gamma}{\alpha_{mul}} \quad (9)$$

$$DF(adr) = \frac{\Delta\alpha_{adr} * \beta_{adr} * \gamma}{\alpha_{adr}} \quad (10)$$

$$DF(reg) = \frac{\Delta\alpha_{reg} * \beta_{reg} * \gamma}{\alpha_{reg}} \quad (11)$$

$$DF(Rclk) = \frac{\Delta A(Rclk) * C_{Rclk}}{N_{Rclk}} \quad (12)$$

PROPOSED METHODOLOGY

B. Execution time parameter

- The dominance factor (DF) for resources R_{mul} , R_{adr} , R_{reg} and R_{clk} for execution time can be obtained similarly as watermark design cost parameter explained in previous section:

$$DF(mul) = \frac{\Delta\alpha_{mul} * T_{mul}}{\alpha_{mul}} * (T_P)^{MAX} \quad (13)$$

$$DF(adr) = \frac{\Delta\alpha_{adr} * T_{adr}}{\alpha_{adr}} * (T_P)^{MAX} \quad (14)$$

$$DF(reg) = \frac{\Delta\alpha_{reg} * T_{reg}}{\alpha_{reg}} * (T_P)^{MAX} \quad (15)$$

- The DF model for the clock oscillator is defined as:

$$DF(Rclk) = \frac{T_{Rclk}^{max} - T_{Rclk}^{min}}{N_{Rclk}} \quad (16)$$

PROPOSED METHODOLOGY

C. Watermarking Framework

- To generate desired signature, comprises of watermarking variables i , I , T , $!$, we utilize the following encoding rule:
 - ❑ ' i ' represents the encoded value of storage variable pair (*prime, prime*)
 - ❑ ' I ' represents the encoded value of storage variable pair (*even, even*).
 - ❑ ' T ' represents the encoded value of storage variable pair (*odd, even*).
 - ❑ ' $!$ ' represents the encoded value of storage variable pair (*0, any integer*).

TABLE 1. WATERMARK SIGNATURE

S1=150 Digits occurrences	S2=175 Digits occurrences	S3= 200 Digits occurrences
$i=40$	$i=50$	$i=55$
$I=40$	$I=45$	$I= 55$
$T=40$	$T=45$	$T= 55$
$!=30$	$!=35$	$!= 35$

TABLE 2 GENERATED WATERMARK SECURITY CONSTRAINTS

Signature digits	i (prime, prime)	I (even, even)	T (odd, even)	$!$ (0, any integer)
S1=150	(V2,V3),(V2,V5),....., (V5,V17) (V5,V19)	(V0,V4),(V0,V6),.....,(V2,V14),(V2,V16)	(V1,V2),(V1,V4),.....,(V3,V10),(V3,V12)	(V0,V1),(V0,V2),.....(V0,V55),(V0,V57)
S2=175	(V2,V3),(V2,V5),.....,(V5,V59),(V5,V61)	(V0,V4),(V0,V6),.....,(V2,V16),(V2,V18)	(V1,V2),(V1,V4),.....,(V3,V20),(V3,V22)	(V0,V1),(V0,V2),.....(V0,V34),(V0,V35)
S3=200	(V2,V3),(V2,V5),.....,(V7,V17), (V7,V19)	(V0,V4),(V0,V6),.....,(V2,V44),(V2,V46)	(V1,V2),(V1,V4),.....,(V3,V40),(V3,V42)	(V0,V1),(V0,V2),.....,(V0,V34), (V0,V35).

DEMONSTRATION OF THE PROPOSED EXPLORATION APPROACH FOR CONVOLUTION LAYER WATERMARK IP: *A. Watermark Design Cost parameter*

TABLE 3. SYSTEM SPECIFICATIONS AND CONSTRAINTS

1.	Watermark design cost constraints (Wc): 18988 units / Area constraints :1900 area units
2.	Maximum time of execution (Tc): 102.4 us (for D =64x64 pixels of input image)
3.	Maximum resources available for the system design: a) 2 Adder units b) 18 Multiplier units c) 3 clock frequency oscillators: 24 MHz, 100 MHz and 400 MHz
4.	No. of clock cycles needed for multiplier and adder/subtractor to finish each operation: 4cc and 2cc
5.	Area occupied by each adder, register and multiplier: 12 area units (a.u), 10 a.u., and 65a.u on the chip (e.g. 12 CLBs on FPGA for adder)
6.	Area occupied by the 24 MHz, 100 MHz and 400 MHz clock oscillator: 6 a.u., 10 a.u. and 14 a.u.
7.	Cost per area unit resource (γ) = 10 units and Cost per area unit clock oscillator = 8 units

- Calculation of the dominance factor of various resources such as multipliers (mul), adder (adr), register (reg), and clock (Rclk) for watermark design cost is shown below.

$$DF(mul) = \frac{\Delta\alpha_{mul}*\beta_{mul}*\gamma}{\alpha_{mul}} = \frac{(18-1)*65*10}{18} = 613.88 \quad (17)$$

$$DF(adr) = \frac{\Delta\alpha_{adr}*\beta_{adr}*\gamma}{\alpha_{adr}} = \frac{(2-1)*12*10}{2} = 60 \quad (18)$$

$$DF(reg) = \frac{\Delta\alpha_{reg}*\beta_{reg}*\gamma}{\alpha_{reg}} = \frac{(70-36)*10*10}{70} = 48.57 \quad (19)$$

$$DF(Rclk) = \frac{\Delta A(Rclk)*C_{Rclk}}{N_{Rclk}} = \frac{(14-6)*8}{3} = 21.33 \quad (20)$$

DEMONSTRATION OF THE PROPOSED EXPLORATION APPROACH FOR CONVOLUTION LAYER WATERMARK IP

B. Demonstrating Watermark Embedding on Convolution Layer IP Design for Design Cost Computation

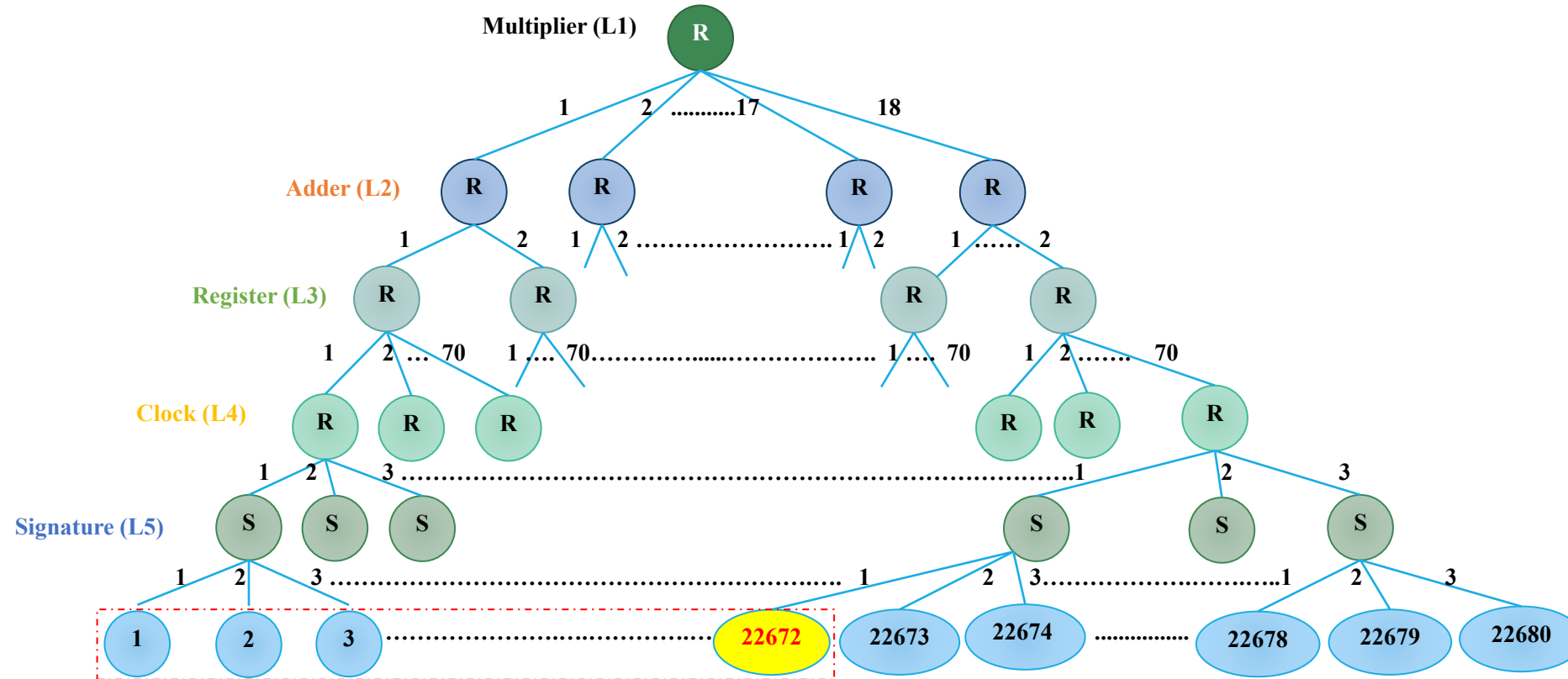


Fig. 4. Tree architecture representing cost arranged in increasing order

B. Demonstrating Watermark Embedding on Convolution Layer IP Design for Design Cost Computation

- The convolutional layer function in CNN that is mathematically represented

$$O_y = \sum_{U,u=lower\ value}^{U,u=upper\ value} \left(\sum_{V,v=lower\ value}^{V,v=upper\ value} I_{UV} X Z_{UV} \right) \quad (21)$$

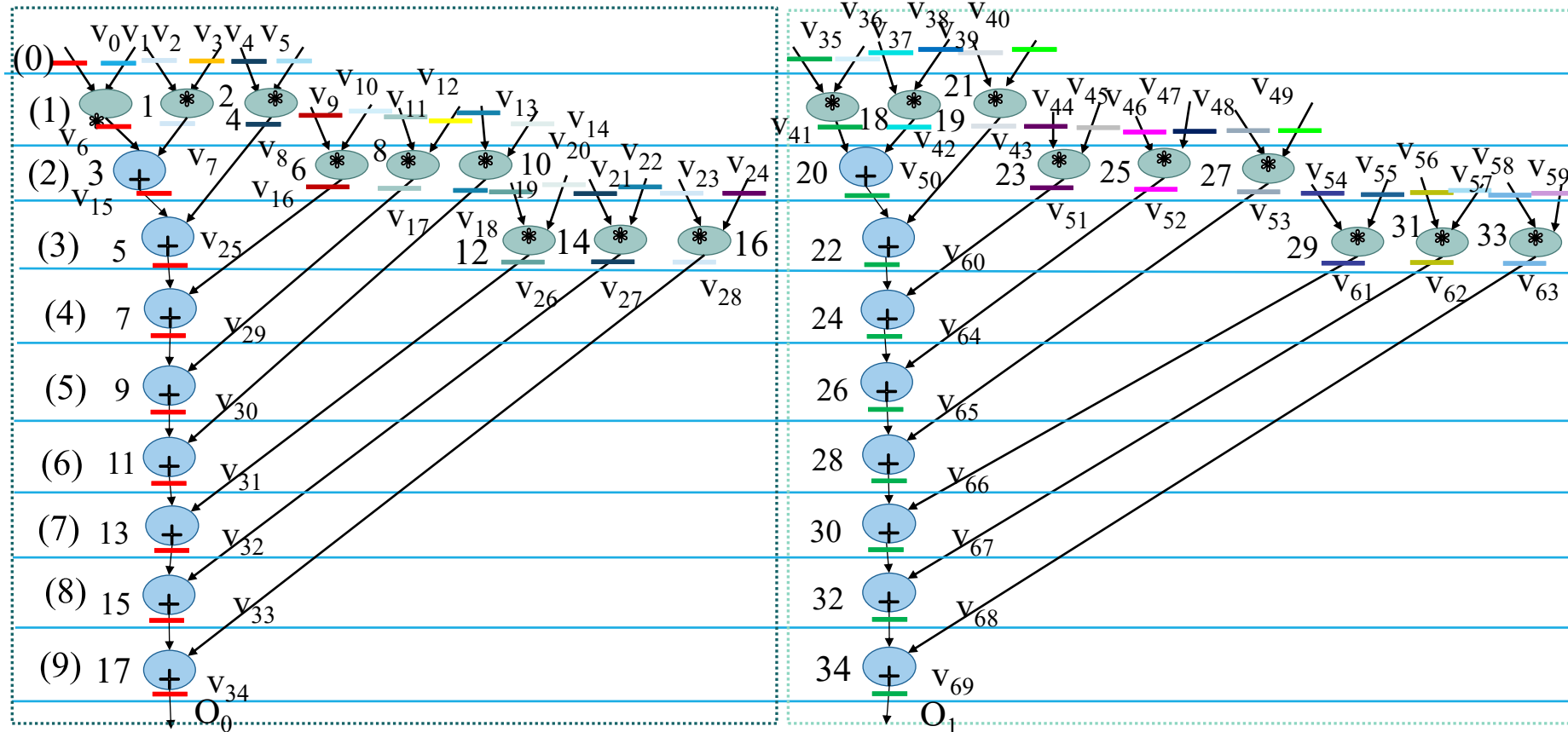


Fig. 5. Scheduled data flow graph of loop unrolled CNN using 6M and 2A

B. Demonstrating Watermark Embedding on Convolution Layer IP Design for Design Cost Computation

[illegible]

C. Execution Time Parameter

- Calculation of the dominance factor of various resources for execution time

$$DF(mul) = \frac{(18-1)*4}{18} * (0.0416) = 0.157 \quad (22)$$

$$DF(adr) = \frac{(2-1)*2}{2} * (0.0416) = 0.041 \quad (23)$$

$$DF(reg) = \frac{(70-1)*2}{70} * (0.0416) = 0.082 \quad (24)$$

$$DF(Rclk) = \frac{6314.66 - 102.4}{3} = 2070.75 \quad (25)$$

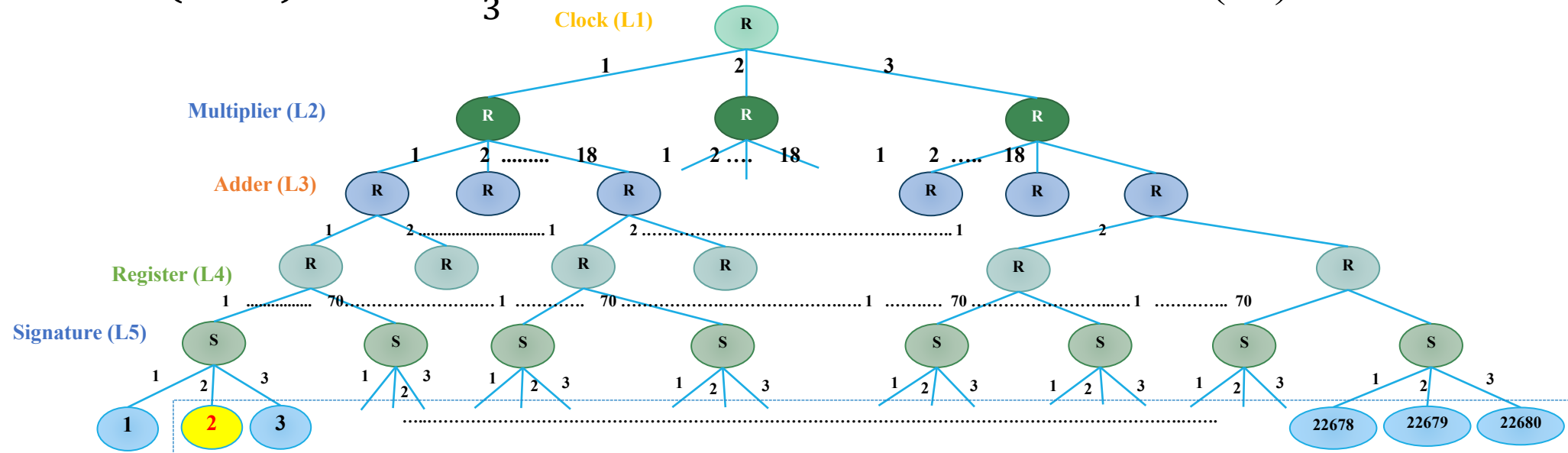


Fig. 6. Tree architecture representing execution time arranged in decreasing order

RESULT AND ANALYSIS

➤ Tamper Tolerance :

$$TT = (V)^r$$

Where, V and r corresponds to types of watermark signature bits and generated watermarking strength of the corresponding security approach.

➤ Probability of Coincidence :

$$PC = \left(1 - \frac{1}{R}\right)^w$$

Where, 'R' represents the total number of registers prior to the incorporation of hardware security constraints and 'w' signifies the total quantity of watermarking constraints integrated into the design.

➤ Design cost :

$$C_f = 0.5 \left(\frac{T_{exe}}{T_{Max}} \right) + 0.5 \left(\frac{A}{A_{Max}} \right)$$

Where, 'T_{exe}' represents the execution time of solution, 'T_{max}' represents maximum execution time, 'A' represents watermark design area, and 'A_{max}' represents maximum area of watermark design.

RESULT AND ANALYSIS

TABLE 6.SPEEDUP EVALUATION OF THE PROPOSED APPROACH FOR WATERMARKED IPs

Watermarked applications	Available Watermark Signature Strength	Available Clock Osc.	Exhaustive variances in the design space	Evaluation of variances using the proposed approach			Speed up wrt exhaustive analysis
				Cost	Execution time	Total	
CNNType-1,UF=2 (18M, 2A, CLK=3,reg=70,WS=3)	S1=150 digits,S2= 175 digits,S3= 200 digits	Clk Osc1= 24MHz	22680	17	17	34	99.92%
CNN Type-2, UF=2 (18M, 2A, CLK=3,reg=70,WS=2)	S1=150 digits.S3= 200 digits	Clk Osc2= 100 MHz	15120	16	16	32	99.89%
CNN Type-3, UF=1 (9M, 1A,CLK=3,reg=36,WS=3)	S1=150 digits,S2= 175 digits,S3= 200 digits		2916	14	14	28	99.51%
CNN Type-4, UF=1 (9M, 1A, CLK=3, reg=36,WS=2)	S1=150 digits,S3= 200 digits	Clk Osc3= 400 MHz	1944	13	13	26	99.33%

TABLE 5. MAX AND MIN EVALUATION FOR COST AND EXECUTION TIME PARAMETERS OF PROPOSED APPROACH

Watermarked applications	Cost		Execution time	
	MIN	MAX	MIN	MAX
CNN Type-1, UF=2 (18M,2A,CLK=3,Reg=70,WS=3)	443 units	1908 units	102.4 us	6314.66 us
CNN Type-2, UF=2 (18M,2A,CLK=3,Reg=70,WS=2)	443 units	1908 units	102.4 us	6314.66 us
CNN Type-3, UF=1 (9M,1A, CLK=3,Reg=36,WS=3)	443 units	971 units	102.4 us	3242.66 us
CNN Type-4, UF=1 (9M,1A,CLK=3,Reg=36,WS=2)	443 units	971 units	102.4 us	3242.66 us

TABLE 7.DESIGN COST, Pc AND TT OF THE PROPOSED APPROACH

Watermarked applications	Register Pre embedding	Register Post embedding	Watermark ed Design Cost	Pc	TT
CNN Type-1	70	70	0.97	5.62E-2	2.58E+120
CNN Type-2	70	70	0.97	8.06E-2	2.29E+105
CNN Type-3	36	36	1	3.57E-3	2.58E+120
CNN Type-4	36	36	1	7.22E-3	2.29E+105

REFERENCE

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Thank You!